

A reporting checklist for large language models in behavioural science

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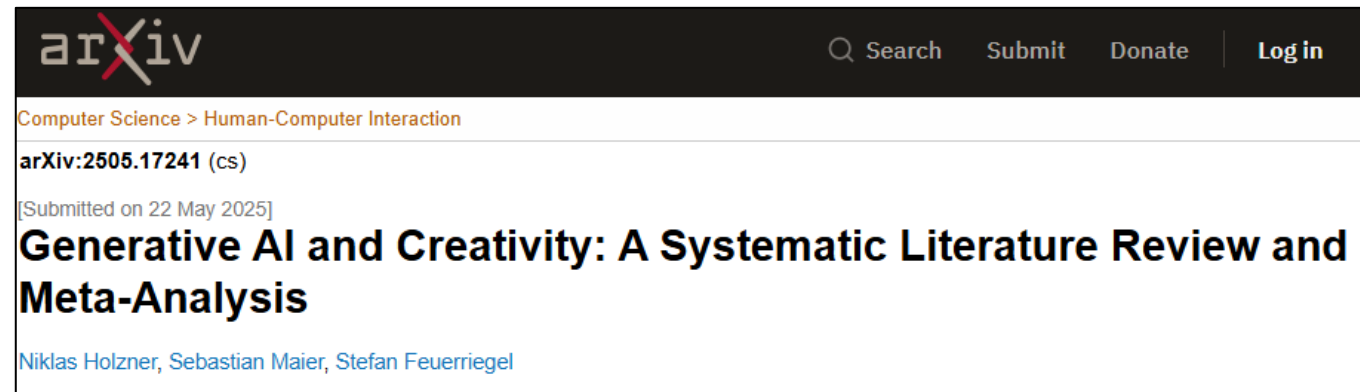
A reporting checklist for large language models in behavioural science

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Considerable research opportunities with LLMs

Increasing use of LLMs for social sciences

- LLMs as research object
 - Digital twins
 - Player in games
 - ...
- LLMs as research tools
 - Data labeling
 - Natural language analysis
 - ...



Group exercise (in teams of 2-4)

You have done a nice research project on how effective LLMs are in innovation management. What things do you report about LLM use?



Goal: shift in reporting

“ We used ChatGPT to generate experimental stimuli. Temperature was set to 0.7 ”

Cannot be reproduced:

- Unknown version
- Unknown system prompt
- Unknown access mode

“ We used gpt-4o-2024-11-20 accessed via API in separate calls (no persistent memory). Temperature: 0.7, Seed: 42. Verbatim prompts and Python scripts are available in the OSF repository.” ”

Methodologically clear:

- Fully reproducible architecture
- Transparent parameters
- Verifiable logic

Unobserved heterogeneities in LLM use

Use of LLMs has many degrees of freedom

- Parameters
- Model version
- Context engineering
- ...

Additional issues

- Data leakage
- No information about training / fine-tuning data
- Biases
- ...

No standardized reporting of how LLMs were used, raising issues of

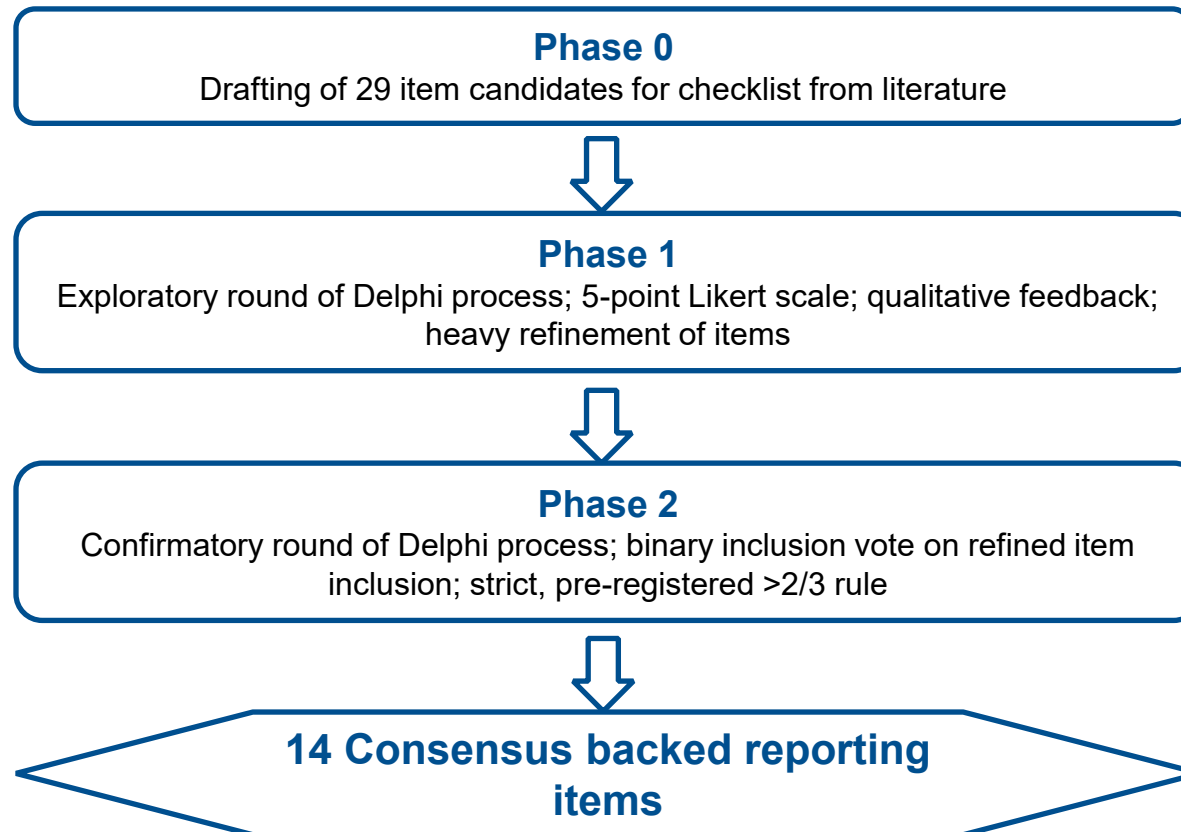
- Reproducibility
- Transparency
- Accountability

GUIDE-LLM as a **reporting checklist** for research involving
LLMs in the **social and behavioral sciences**

Development of GUIDE-LLM

Delphi process with a global panel of researchers across disciplines (**N=80**)

- Structured way to gather expert opinions to identify (dis)agreement and whether a reasonable consensus can be reached



7 Dimensions, 14 items

Dimension 1: Scope of LLM use

Explaining how exactly LLMs were used

LLMs were used in this project for...

The degree of automation (HitL v. automated)

Dimension 2: LLM specifics

Details on what system was used exactly

Model name, provider, model size, access date, link, ...

Model access (e.g., API) and context mode (e.g., chat)

Relevant model configs, e.g., seed, runs, temperature, ...

Customization: ...

Persistence of memory across sessions?

Dimension 3: Prompts

What were model instructions

Exact user prompts: ...

Exact system prompts (if any): ...

Dimension 4: Data inputs & privacy

Sensitivity of input data

Handling of personal or sensitive input data (if any)

7 Dimensions, 14 items

Dimension 5: Validation & Interpretation

How was output of LLM handled

Human validation of LLM output

Description of relevant post-processing (e.g., filtering)

Dimension 6: Reproducibility

How can other researcher reproduce results

Code/notebooks/... for LLM use shared on ...

Dimension 7: Competing interests

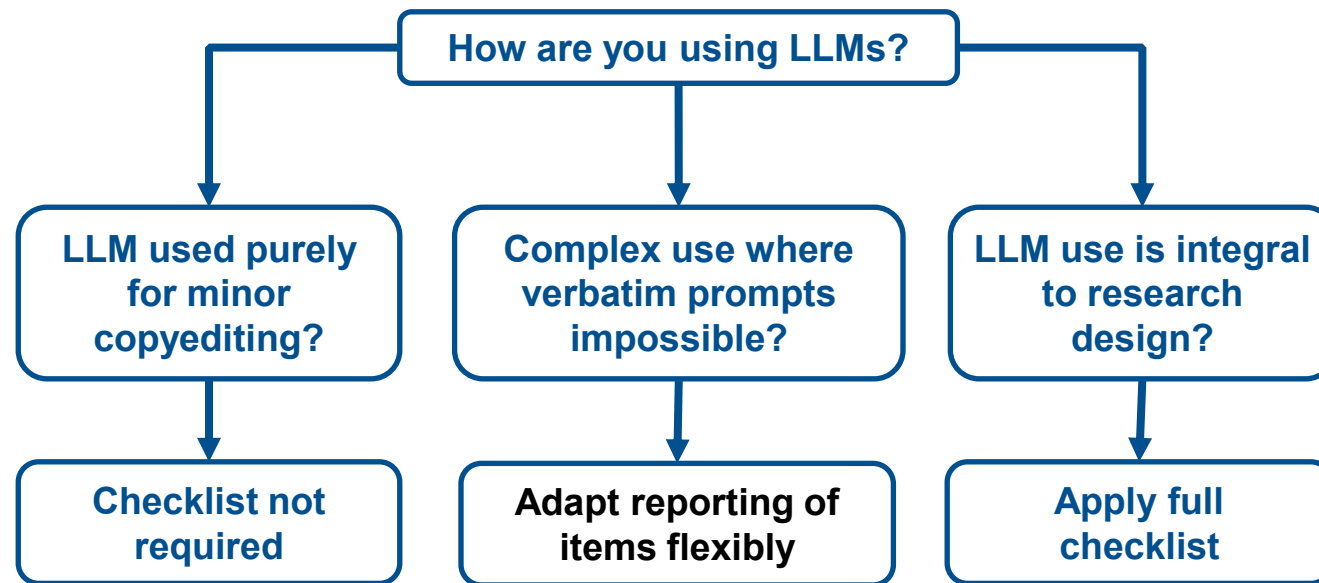
Connection to major technology provider

Funding, support, ... (e.g., in-kind access to compute): ...

Practical implementation

GUIDE-LLM is a pragmatic framework, not a bureaucratic straightjacket

- Simplify reporting by cross-referencing existing manuscript sections
- Idea of a living document



<https://www.llm-checklist.com/>

How to use this checklist

For a step-by-step guide and examples of real research workflows:

 [How to use GUIDE-LLM](#)

For background on the motivation and principles behind the checklist, see

 [Why GUIDE-LLM?](#)

Access the associated paper published in *Nature Human Behaviour*:

 [Paper with detailed instructions](#)

Download the checklist

- [Download DOCX](#)
- [Download LaTeX](#)

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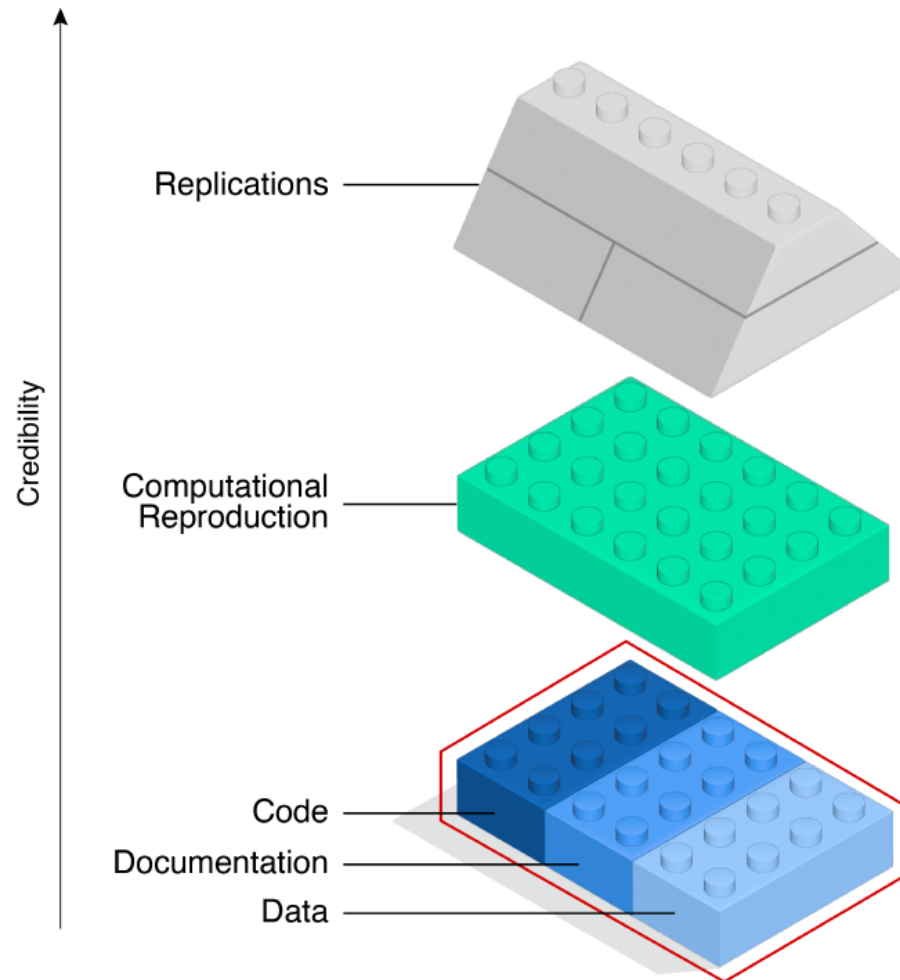
Improving computational reproducibility in the social sciences

[Michael V. Reiss](#), [Susanne J. Adler](#), [Christopher Barrie](#), [Jana B. Berkessel](#), [Johannes Breuer](#), [Abel Brodeur](#), [Garret Christensen](#), [Tobias Dienlin](#), [Jeremy Freese](#), [Ben Greiner](#), [Mario Haim](#), [Tom E. Hardwicke](#), [Magnus Johannesson](#), [Gary King](#), [Frauke Kreuter](#), [Christopher Lucas](#), [Ana Martinovici](#), [Christophe Pérignon](#), [Marko Sarstedt](#), [Felix D. Schönbrodt](#), [Oleg Urminsky](#), [Wouter van Atteveldt](#), [Simine Vazire](#), [Lars Vilhuber](#), [Florian von Wangenheim](#), [Eric-Jan Wagenmakers](#), [Nicolas Pröllochs](#), [Claire Robertson](#), [Felix Holzmeister](#), [Stefan Feuerriegel](#) & [Hauke Roggenkamp](#) 

Nature Human Behaviour (2026)

Code underlying published findings in the social sciences often fails to reproduce reported results when others re-run it on the original data. This Comment proposes four recommendations to strengthen computational reproducibility, facilitating the trustworthiness and reliability of research.

Computational reproducibility as a foundation of scientific credibility



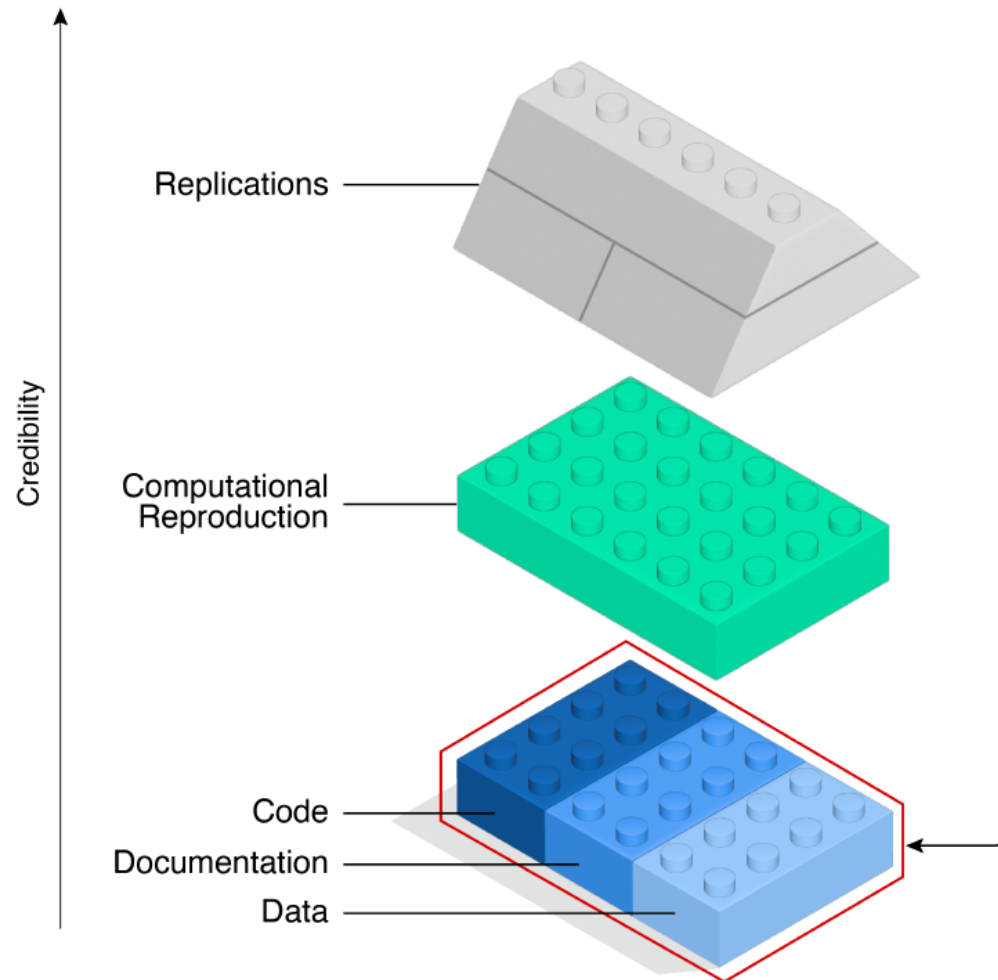
Group exercise (in teams of 2-4)

What are best practices to support computational reproducibility (in your lab, in your community, in your own view)?

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Computational reproducibility as a foundation of scientific credibility



Four recommendations to improve computational reproducibility:

Script. Orchestrate the entire analysis via code. Use a main or "controller" script to eliminate manual interventions and ensure a deterministic path from raw data to final figures.

Encapsulate. Specify the exact computational environment to promote portability. Record all software versions and dependencies. Use relative paths, standard folder structures and environment managers or containerization.

Explain Create human-readable documentation that clarifies the project structure and explains how to access any data. Include data dictionaries and code comments to facilitate peer review and future reuse.

Deposit. Share code, data, and documentation in persistent, public repositories (e.g., Dataverse, OSF, Zenodo). Adhere to FAIR principles while ensuring sensitive data is appropriately de-identified and restricted.

SCRIPT

Automate analysis pipeline

Script all analysis steps from data retrieval to final analysis results, eliminate GUI interactions that leave no record, and provide a single entry point that runs the full pipeline in order (e.g., a `run_all.R` or `main.py` that sources all scripts in sequence). Tools like [Snakemake](#) can assist managing such pipelines.

Set random seeds

Set and document a seed in every script that uses randomization (e.g., `set.seed(20261305)` in R, `np.random.seed(20261305)` in Python, placed at the top of each script).

Protect raw data

Store raw data in a dedicated read-only folder and save all processed versions separately (e.g., originals in `data/raw/` and cleaned data in `data/processed/`).

Keep intermediate outputs

Save computationally expensive intermediate outputs so downstream scripts can load them.

ENCAPSULATE

Minimise dependencies

Add a package only when it provides clear value over base language functions (prefer base R or the standard library in Python for simple operations).

Use relative paths

Use paths [relative to the project root](#) and avoid OS-specific commands.

Capture and document the environment

Record all package and language versions in a lockfile and note the OS and critical dependencies in the README. Use `groundhog` or `renv` in R or `conda/pip` in Python.

Containerise when warranted

For complex projects, provide a container capturing the full system environment⁶. Docker bundles code and system libraries into a single portable image.

Test before sharing

Run all code in a fresh environment before sharing (ask a colleague to reproduce your results without further instructions, or use a cloud instance such as [Binder](#), Google [Colab](#), or Posit [Cloud](#)).

EXPLAIN

Write a README

Create a human-readable file, for example by completing the Social Science Data Editors' [template README](#). State how each dataset can be accessed (in particular for confidential or restricted data), its provenance (sources, collection or retrieval method, dates, and licensing), the software and hardware requirements, environment setup, execution order, and expected runtime. See also Figure 4 in ref. 7.

Link code to manuscript

Add comments indicating which outputs correspond to tables, figures, and other elements in your paper (e.g., `# Creates Table 1`, placed directly above the relevant code block).

Create a data dictionary

Document each variable's name, type, values, and units (e.g., `age_years`, numeric, 18–65, years). Follow field-specific metadata standards.

Write readable code and comment decisions

Use variable names that make sense to humans. Comment and explain non-obvious choices, transformations, and exclusion criteria (e.g., `# Log-transform: right-skewed distribution; # Exclude id=047: failed pre-registered attention check`).

DEPOSIT

Organize for others	Confirm that the folder structure is intuitive and follows field-specific standards, if available, that file names reflect their contents and position in the workflow, and that variable names are descriptive and consistent for someone unfamiliar with the project.
Deposit in a persistent repository	Archive code, data, and documentation in a repository that issues a persistent DOI (e.g., Dataverse, OSF, or Zenodo; see also re3data.org).
Handle restricted data	When data cannot be shared publicly, document how others can access or reproduce them, arrange verifier access at the design stage (e.g., in data-sharing agreements or ethics committee applications), consider data enclaves, non-consumptive environments, or synthetic data, and include log files from code runs on the original data. Share analysis code regardless.
Declare license	Add a LICENSE file specifying terms of reuse for code and data separately (see choosealicense.com for code licenses, e.g., MIT; and creativecommons.org for data licenses, e.g., CC-BY-4.0).

My personal takeaways

- Reproducibility practice varies across fields (traditional behavioral science vs. fast-moving AI/ML research)
- As an Area Chair at NeurIPS, KDD, etc. => **always make your code available** (from round 1). Reviewers are not forgiving
- **README files**: overlooked but, if checked, often problematic
=> https://social-science-data-editors.github.io/template_README/template-README.html
 - Include short how-to-use, dependencies, etc.

Using natural language processing to analyse text data in behavioural science

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